How much of the energy is actually available?

? Two blocks collide. The total kinetic energy is 100 J, but only part of it can become heat or deformation. Which part?

1 ## Two pieces of one number

Write each velocity as the centre-of-mass velocity plus a velocity relative to it. The cross terms cancel because the total momentum in the CM frame is zero, and the kinetic energy separates into two independent pieces.

$$K = \tfrac{1}{2}Mv_{\text{cm}}^2 + K_{\text{rel}}, \qquad M = m_1 + m_2$$

2 ## The internal part is the reducible one

No internal force can change the CM velocity, so the drift energy is untouchable from inside. Everything a collision, an explosion or a spring can convert comes from the relative part, and that is why the reduced mass appears in every collision formula.

$$K_{\text{rel}} = \tfrac{1}{2}\mu v_{\text{rel}}^2, \qquad \mu = \frac{m_1 m_2}{m_1 + m_2}$$

→ a perfectly inelastic collision converts exactly K_rel into heat

3 ## The two limits to remember

Equal masses moving toward each other at the same speed: the CM is at rest and all the energy is internal, so all of it can be lost. A light particle hitting a heavy one at rest: almost all the energy is drift, so very little can be converted, which is why a bullet heats a wall far more easily than a ping-pong ball heats a block.

$$m_1 = m_2 : \; K_{\text{rel}} = \tfrac{1}{2}K_{\text{tot}}, \qquad m_1 \ll m_2 : \; K_{\text{rel}} \approx \tfrac{m_1}{m_2}K_{\text{tot}}$$

4 ## A worked split

A 3 kg block at 4 m/s hits a 1 kg block at rest and they stick. Find the CM speed, the CM energy, and the energy lost.

$$V = \frac{12}{4} = 3 \text{ m/s}, \qquad K_{\text{cm}} = \tfrac{1}{2}(4)(9) = 18 \text{ J}$$

$$K_i = 24 \text{ J} \;\Rightarrow\; \text{lost } 6 \text{ J}, \;\; \text{which is } K_{\text{rel}}$$

→ the 18 J of drift survives the collision untouched

KEY RESULT

the decomposition

$$K = \tfrac{1}{2}Mv_{\text{cm}}^2 + \tfrac{1}{2}\mu v_{\text{rel}}^2$$

NUMBERS

a check

2 kg at 3 m/s meets 1 kg at rest

mu = 2/3, K_rel = 3 J of the 9 J total

INSIGHT

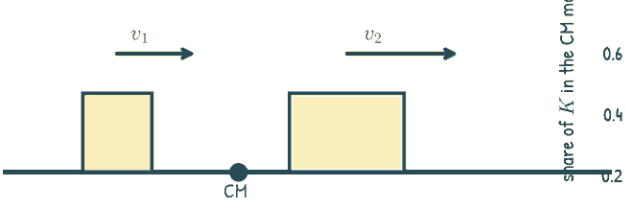
why reduced mass appears

It is the mass that carries the internal motion. Every two-body collision formula is a one-body formula with mu in place of m.

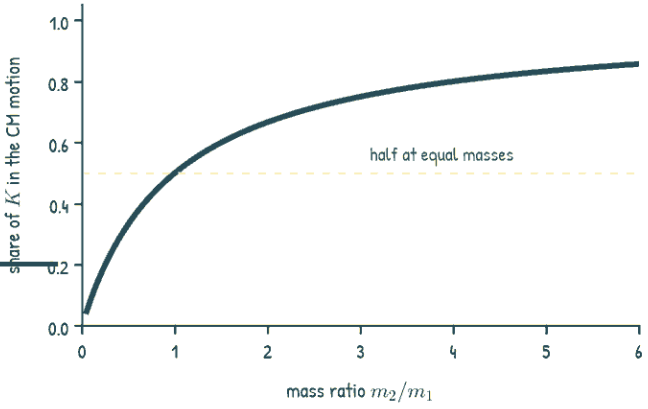
cm_decomposition

split the motion into drift plus internal motion

$$K = \tfrac{1}{2}Mv_{\text{cm}}^2 + \tfrac{1}{2}\mu v_{\text{rel}}^2$$



the drift usually carries most of it



KEEP / split the kinetic energy into drift of the centre of mass plus internal motion

Why does angular momentum keep a planet out of the Sun?

? Gravity pulls inward with nothing to oppose it. Why does a planet not simply fall in?

1 ## Separate the two motions

In a central field the angular momentum is constant, so the angular part of the kinetic energy is fixed by the radius alone. Move it to the other side of the energy equation and it behaves like an extra potential.

$$E = \tfrac{1}{2}m\dot{r}^2 + U_{\text{eff}}(r), \qquad U_{\text{eff}} = U(r) + \frac{L^2}{2mr^2}$$

2 ## Read the effective curve

The added term is positive and falls as 1/r squared, so it dominates at small radii and creates a minimum when combined with an attractive potential. The minimum is the circular orbit; a small radial oscillation around it is an ellipse.

$$\frac{dU_{\text{eff}}}{dr} = 0 \;\Rightarrow\; r_0 = \frac{L^2}{GMm^2}$$

→ no force pushes outward: the barrier is the cost of conserving angular momentum

3 ## Where it is genuinely useful

The same construction gives the critical angular momentum below which a particle falls in, the turning points of a comet, and the stability condition for a circular orbit. It also explains the bead on a rotating hoop (OLY 08), which is the same trick in a rotating frame.

4 ## A worked barrier

A satellite orbits at 7000 km from the Earth centre. Find the effective potential at that radius per unit mass, and compare the two terms.

$$-\frac{GM}{r} = -5.69 \times 10^{7}\ \text{J/kg}$$

$$+\frac{v^2}{2} = +2.85 \times 10^{7}\ \text{J/kg}$$

→ the sum is minus half the magnitude: the virial relation for a circular orbit

KEY RESULT

effective potential

$$U_{\text{eff}} = U(r) + \frac{L^2}{2mr^2}$$

the angular part of K, rewritten as a barrier

NUMBERS

a check

Earth: L = 2.66e40, GMm² gives r = 1.5e11 m

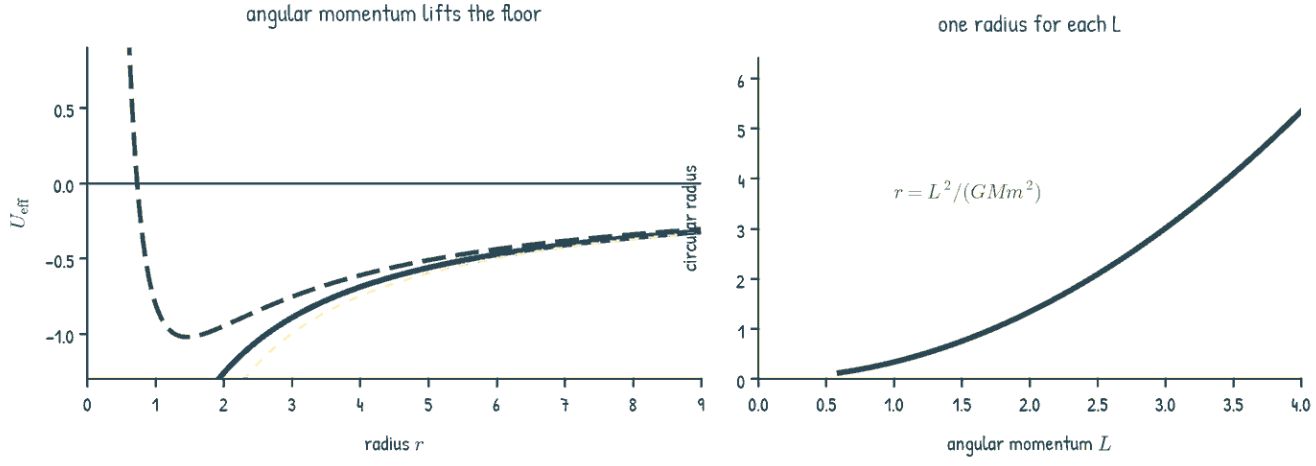
the minimum of U_eff reproduces the orbit radius

TRAP

calling it a centrifugal force

In an inertial frame there is no outward force. The barrier is energy bookkeeping; the only force is inward.

effective_potential



KEEP / the angular term is a repulsive barrier made of kinetic energy, not a force

How far does a bouncing ball travel before it stops?

? A ball is dropped from h and returns to e²h after each bounce. The bounces are infinite; is the total distance infinite too?

1

Write the two series

The first fall is h. Every later bounce contributes a rise and a fall, each shorter by e squared. The flight time scales by e rather than e squared, because time goes as the square root of height.

$$s = h\frac{1+e^2}{1-e^2}, \qquad t = \sqrt{\frac{2h}{g}} \cdot \frac{1+e}{1-e}$$

2

The energy view

Each impact multiplies the speed by e and therefore the kinetic energy by e squared. The energy lost per bounce is a fixed fraction of what is left, so the sequence is geometric and the total lost before rest is exactly the initial mgh.

$$E_n = e^{2n}mgh, \qquad \sum E_{\text{lost}} = mgh$$

→ the ball never gets negative energy: the sum is bounded by the initial store

3

Why the time is finite while the bounces are not

The bounce count diverges but the durations shrink faster than the count grows. The series converges to a finite time, so the ball comes to rest in finite time after infinitely many bounces, which is the standard counterexample to "infinitely many events need infinite time".

$$t_0(1 + 2e + 2e^2 + \ \dots\ ) = t_0\frac{1+e}{1-e}$$

4

A worked bounce

A ball dropped from 1.8 m rebounds to 1.15 m. Find e, the total distance and the total time.

$$e = \sqrt{1.15/1.8} = 0.80, \qquad s = 1.8\frac{1.64}{0.36} = 8.2\ \text{m}$$

$$t = \sqrt{0.367} \cdot 9 = 0.606(9) = 5.45\ \text{s}$$

→ one bounce is over in 0.6 s; the infinite tail takes five more

KEY RESULT

the closed forms

$$s = h\frac{1+e^2}{1-e^2}, \qquad t = \sqrt{\frac{2h}{g}} \cdot \frac{1+e}{1-e}$$

NUMBERS

a check

h = 10 m, e = 0.5

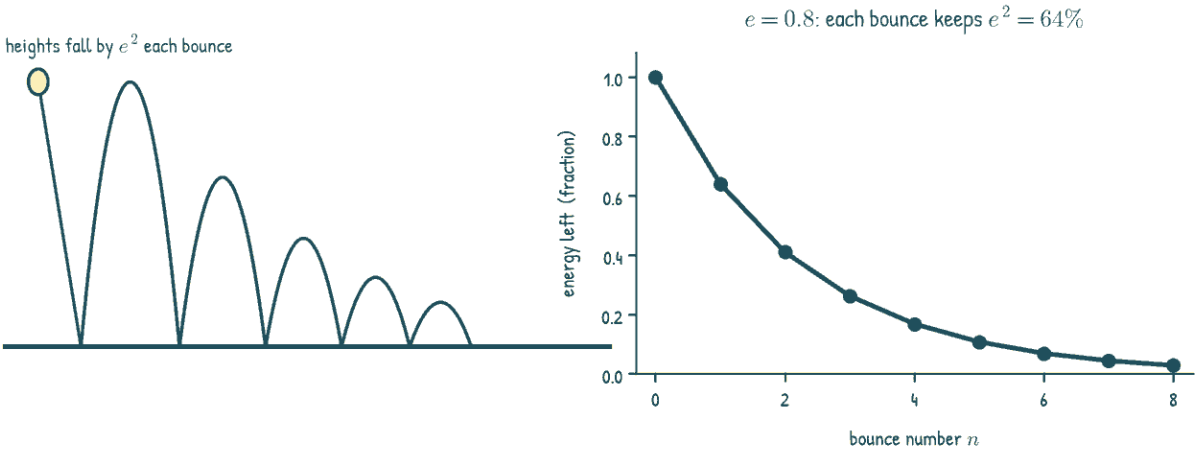
s = 16.7 m, t = 4.29 s

TRAP

the factor of two

Every bounce after the first contributes twice its height, once up and once down.

bounce_energy



KEEP / two geometric series, one for distance and one for time, both with finite sums

Why does a rolling ball beat a sliding block downhill?

? A ball and a block are released from the same height on the same slope. The block has no friction, the ball rolls. Which arrives first?

1 Two kinetic terms

A rolling body translates and rotates at once. With the rolling condition the rotation is fixed by the speed, so the total kinetic energy is a fixed multiple of half m v squared.

$$K = \tfrac{1}{2}mv^2 + \tfrac{1}{2}I\omega^2 = \tfrac{1}{2}mv^2\left(1 + \tfrac{I}{mR^2}\right)$$

2 The shape sets the multiple

A ring has I = mR squared, so it carries twice the energy of a sliding block at the same speed and is the slowest. A disc gives three quarters, a solid sphere seven tenths. The order of arrival downhill is always sphere, disc, ring, regardless of mass and radius.

$$K:\ \tfrac{1}{2}mv^2\ (\text{slide}),\ \tfrac{3}{4}mv^2\ (\text{disc}),\ \tfrac{7}{10}mv^2\ (\text{sphere}),\ mv^2\ (\text{ring})$$

→ the ratio I/mR squared is the only thing that matters

3 Where the energy goes on a slope

Energy conservation from height h gives the speed at the bottom with the same factor in the denominator. Static friction does the work that spins the body up, but it does no net work on the rolling body because the contact point does not slip.

$$v^2 = \frac{2gh}{1 + I/(mR^2)}$$

4 A worked roll

A solid sphere, a disc and a ring roll from rest down a 2 m high slope. Find each bottom speed.

$$v = \sqrt{\frac{2gh}{1 + I/mR^2}}:\ 5.29,\ 5.11,\ 4.43\ \text{m/s}$$

$$\text{sphere}\ (2/5),\ \text{disc}\ (1/2),\ \text{ring}\ (1)$$

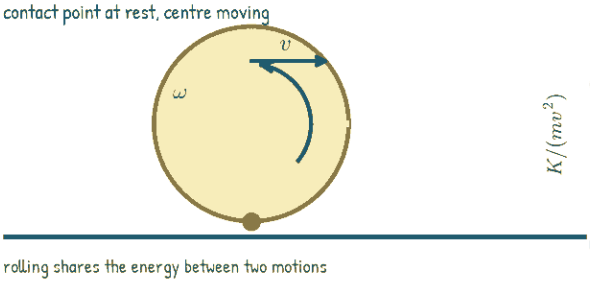
→ all three take the same time only if the slope is long enough to reach these speeds

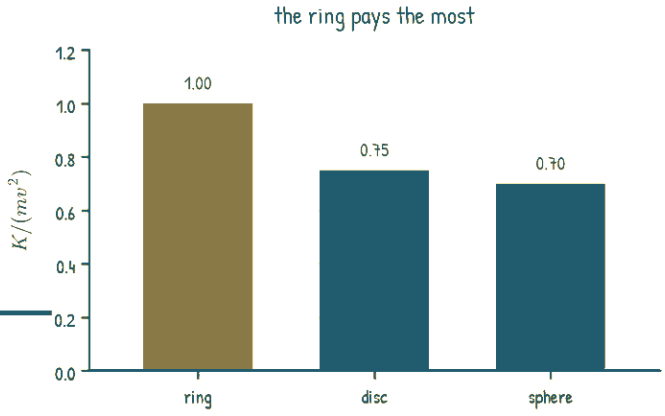
KEY RESULT
rolling energy
$K = \tfrac{1}{2}mv^2\left(1 + \tfrac{I}{mR^2}\right)$

NUMBERS
a check
solid sphere from 1.4 m
v = sqrt(2 x 9.8 x 1.4 / 1.4) = 4.43 m/s

TRAP
adding friction work
Static friction at a non-slipping contact does no work on the rolling body. Do not subtract f d from its energy.

rolling_energy





KEEP / a rolling body spends part of its energy on rotation, so it descends more slowly

Why is the block on a moving wedge slower than sqrt(2gh)?

? A block slides down a smooth wedge that is free to slide on a smooth table. The textbook answer sqrt(2gh) must be wrong. Why?

1

Two unknowns, two equations

Horizontal momentum is conserved for the pair, and mechanical energy is conserved for the pair. Those two equations determine the wedge speed and the relative sliding speed; the block ground speed then follows from the vector sum.

$$u^2 = \frac{2gh(M+m)}{M+m\sin^2\theta}, \qquad V = \frac{mu\cos\theta}{M+m}$$

2

The error the source warns about

Treating u as the block speed is the standard mistake. u is measured along the face relative to the wedge; the ground speed is the vector sum of u and V, and is smaller for a finite wedge.

$$v_b^2 = u^2\left(\sin^2\theta + \left(\frac{M}{M+m}\right)^2\cos^2\theta\right)$$

→ the block ground speed is below sqrt(2gh) for every finite M

3

The two limits

An infinitely heavy wedge cannot recoil and the block recovers sqrt(2gh). A massless wedge carries almost no energy, so the block again approaches the same speed. The deficit peaks when the masses are comparable.

$$M\to\infty:\ v_b\to\sqrt{2gh}, \qquad M\to 0:\ v_b\to\sqrt{2gh}$$

4

A worked wedge

m = 1 kg, M = 4 kg, theta = 30°, released from 0.5 m up the face. Find both speeds at the bottom of the face.

$$u^2 = \frac{2(9.8)(0.5)(5)}{4.25} = 11.5 \ \Rightarrow\ u = 3.39\ \text{m/s}$$

$$V = \frac{1(3.39)(0.866)}{5} = 0.59\ \text{m/s}$$

→ the block drops 0.5 m but its ground speed is only 2.9 m/s, below the 3.1 m/s fixed-wedge value

KEY RESULT

the pair

$$mgh = \tfrac{1}{2}MV^2 + \tfrac{1}{2}mv_b^2$$

plus horizontal momentum conservation

NUMBERS

a check

m = 1, M = 4, theta = 30°, h = 0.5 m

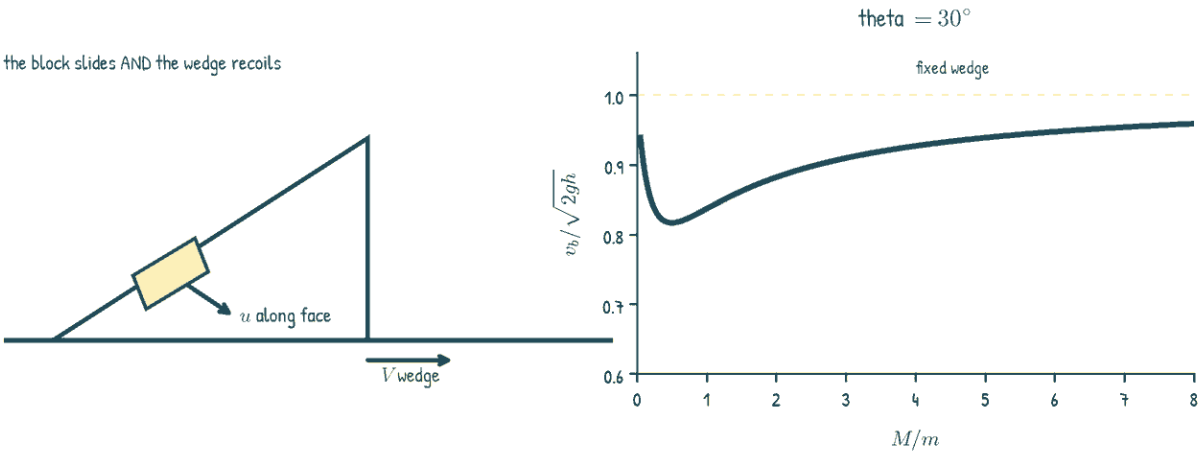
u = 1.60 m/s, V = 0.28 m/s, v_b = 1.15 m/s

TRAP

one-body energy

The block alone does not conserve energy: the normal force does work on it because the face moves.

wedge_energy



KEEP / the wedge takes a share of the energy, so the block cannot have all of it

How much energy does it take to reach orbit?

? Escape needs the full well depth. Orbit needs less, but not as little as you might hope. What fraction?

1

The two energies

The energy at rest on the surface is minus GMm over R. The energy of a circular orbit is minus GMm over twice r, which is the virial relation for a circular orbit. The difference is what the rocket must supply.

$$E_{\text{orbit}} - E_{\text{surface}} = GMm\left(\tfrac{1}{R} - \tfrac{1}{2r}\right)$$

2

The ratio that surprises

For a low orbit, r is close to R and the bracket is about one over twice R. The escape energy is GMm over R. So orbit costs half of escape, and the remaining half is the kinetic energy the satellite keeps in orbit, not a loss.

$$\frac{\Delta E_{\text{orbit}}}{\Delta E_{\text{escape}}} = \frac{1}{2} - \frac{R}{2r}$$

→ the orbital speed is 1/sqrt(2) of the escape speed, which is the same statement

3

Why it matters

Reaching orbit is not mainly about height; it is mostly about the horizontal speed. A sounding rocket that reaches 400 km with no horizontal speed has done far less than a satellite at the same height, which is why launch vehicles pitch over early.

4

A worked launch

Find the energy to lift 1 kg from the Earth surface to a 400 km orbit, and compare it with escape.

$$\Delta E = GM\left(\tfrac{1}{R} - \tfrac{1}{2r}\right) = 6.26 \times 10^7 - 2.94 \times 10^7$$

$$= 3.3 \times 10^7 \ \text{J/kg} = 33 \ \text{MJ/kg}$$

→ escape from the same point is 62.5 MJ/kg, so orbit costs just over half

KEY RESULT

the ratio

$$\Delta E_{\text{orbit}} = \tfrac{GMm}{2R} \ \text{(low orbit)}, \qquad = \tfrac{1}{2}\Delta E_{\text{esc}}$$

NUMBERS

a check

Earth: v_esc = 11.2 km/s, v_orbit = 7.9 km/s

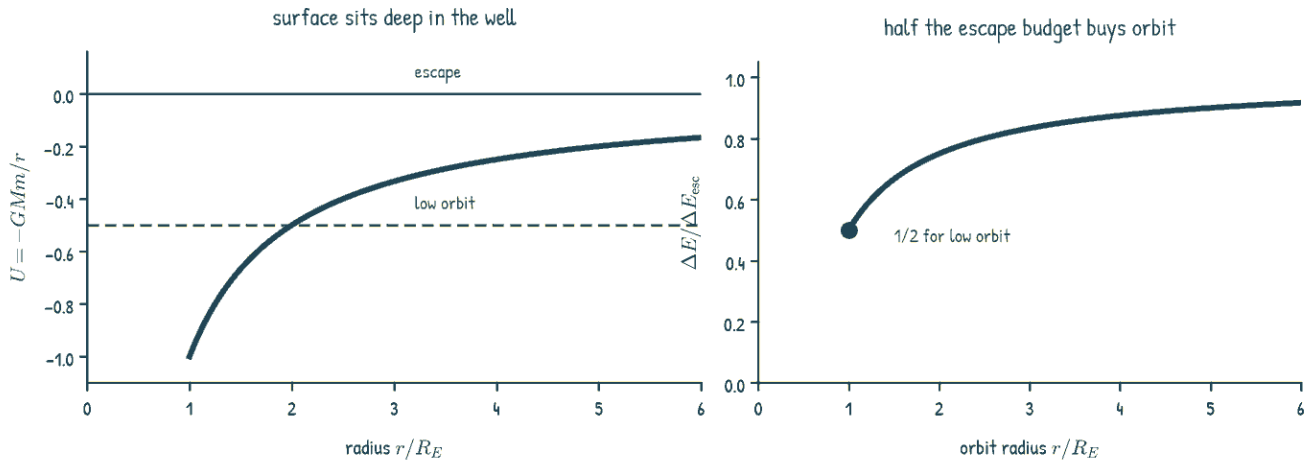
ratio of energies: 1/sqrt(2) squared = 0.5

TRAP

counting the orbital KE as wasted

It is stored, not lost: bringing the satellite down returns it. Only the drag losses are gone.

launch_energy



KEEP / low orbit costs half the escape energy, and the other half is the speed you keep

Where does the missing half of the work go?

? A chain is pulled off a table at constant speed v. The power delivered is F v, but the energy in the chain grows more slowly. Where is the difference?

1 ## The two rates

The force must support the hanging weight and supply the momentum flux. Multiply by v to get the power delivered, and compare it with the rate of change of the mechanical energy.

$$P = Fv = \lambda x g v + \lambda v^3, \qquad \dot{E} = \lambda x g v + \tfrac{1}{2}\lambda v^3$$

2 ## The difference is the collision

Each link is at rest and then shares the speed v of a moving system. That is a perfectly inelastic capture, and half the energy invested in the capture is unavoidably lost, exactly as in any inelastic collision.

$$\dot{Q} = \tfrac{1}{2}\lambda v^3$$

→ the loss scales as v cubed, so fast pulling is disproportionately wasteful

3 ## The same ledger as the rocket

Both the chain and the rocket are open systems: mass crosses the boundary, and the momentum flux must be included. The chain is the inelastic limit of the rocket, with the links being captured instead of expelled.

4 ## A worked unroll

A 4 kg, 2 m chain lies coiled on a table. It is pulled up at a steady 1.5 m/s. Find the pull and the loss rate when 1.2 m is up.

$$F = \lambda x g + \lambda v^2 = 2(9.8)(1.2) + 2(2.25) = 28.0 \text{ N}$$

$$\dot{Q} = \tfrac{1}{2}\lambda v^3 = 3.4 \text{ W of the } 42 \text{ W delivered}$$

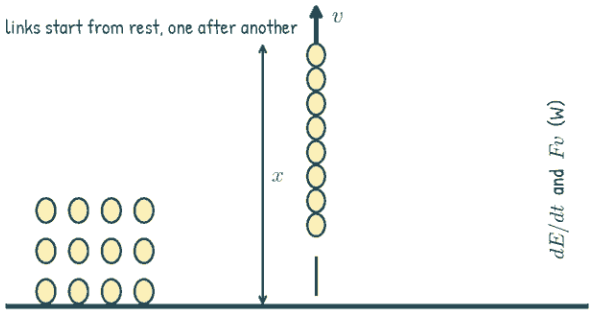
→ 8 per cent is lost in the jerk at this speed, and it grows as v cubed

KEY RESULT
the audit
$$F = \lambda x g + \lambda v^2, \qquad \dot{Q} = \tfrac{1}{2}\lambda v^3$$

NUMBERS
a check
lambda = 2 kg/m, x = 2, v = 1 m/s
F = 41.2 N, loss 1 W of the 41.2 W delivered

INSIGHT
the general rule
Mass captured by a moving system always costs a factor of two: half to motion, half to heat.

chain_unroll





KEEP / links are jerked into motion inelastically, and the lost half is heat

Why does a spinning hoop make the bead climb?

? A bead slides freely on a smooth hoop spinning about its vertical diameter. At high enough speed it leaves the bottom. Where does the energy come from?

1

Two contributions, one curve

Gravity favours the bottom and the rotation favours the equator, because the bead must be further from the axis to have a large radius at high angular speed. Adding the two produces an effective potential whose shape depends on omega.

$$U_{\text{eff}}(\theta) = -mgR\cos\theta - \tfrac{1}{2}m\omega^2R^2\sin^2\theta$$

2

The threshold

Setting the derivative to zero gives the bottom and, for fast enough rotation, a pair of side positions. The condition for the side positions to exist is that the rotational term beats gravity at the bottom.

$$\omega > \omega_c = \sqrt{\frac{g}{R}}, \qquad \cos\theta_0 = \frac{g}{\omega^2 R}$$

→ below the critical speed the only stable seat is the bottom

3

Where the energy comes from

The hoop is driven, so it does work on the bead through the normal force, which has a component along the bead motion once the hoop is rotating. Energy is supplied by the motor, and the bead settles where the effective potential is lowest.

4

A worked hoop

A bead on a 0.4 m hoop which spins at 5 rad/s. Find the critical speed and the equilibrium angle.

$$\omega_c = \sqrt{9.8/0.4} = 4.95 \text{ rad/s}$$

$$\cos\theta = \frac{9.8}{25(0.4)} = 0.98 \;\Rightarrow\; \theta = 11^\circ$$

→ just above the threshold the bead sits very close to the bottom

KEY RESULT

the result

$$\omega_c = \sqrt{g/R}, \qquad \cos\theta_0 = g/(\omega^2 R)$$

NUMBERS

a check

R = 0.5 m, omega = 6 rad/s

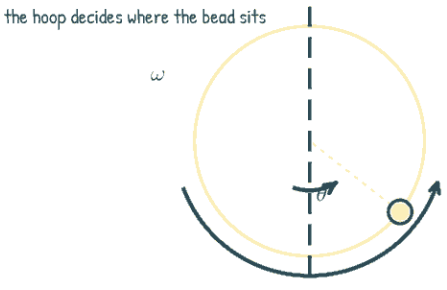
omega_c = 4.43, theta = 57°

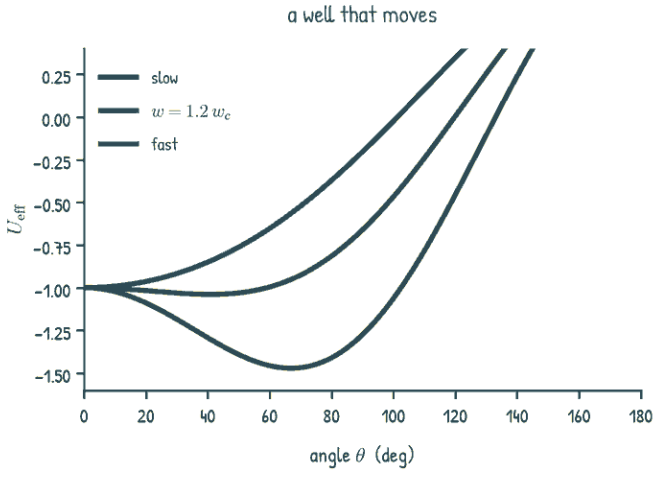
TRAP

energy conservation in the lab frame

The bead alone does not conserve energy. Use the effective potential, or include the work done by the hoop.

bead_energy





KEEP / the hoop does work on the bead, and in the rotating frame that is one effective potential

What does the coefficient of restitution actually measure?

? Two balls drop from the same height onto the same floor and bounce to different heights. What single number separates them?

1 Define it on speeds

The coefficient of restitution is the ratio of separation speed to approach speed along the normal. It is a property of the pair of surfaces and the impact speed, not of the mass.

$$e = \frac{v_{\text{sep}}}{v_{\text{app}}}, \qquad 0 \le e \le 1$$

2 What it does to energy

Since energy goes as speed squared, one bounce multiplies the mechanical energy by e squared. That is why the rebound height is e squared times the drop height, and why a bouncy ball is specified by its rebound ratio rather than by a force.

$$\frac{h_{n+1}}{h_n} = e^2, \qquad \frac{K_{n+1}}{K_n} = e^2$$

→ e is a ratio of speeds; the energy ratio is its square

3 The two extremes

e = 1 is elastic: no energy lost, the ball returns to the same height for ever. e = 0 is perfectly inelastic: the ball stays on the floor, and all the kinetic energy becomes heat and sound. Real balls sit between 0.6 and 0.9.

→ a superball at e = 0.9 keeps 81 per cent of its energy per bounce

4 A worked restitution

A ball falls 3 m and rebounds to 1.08 m. Find e and the fraction of energy kept.

$$e = \sqrt{1.08/3} = 0.60$$

$$\frac{K_f}{K_i} = e^2 = 0.36$$

→ it loses nearly two thirds of its energy at one impact

DEFINE

restitution

$e = v_{\text{sep}}/v_{\text{app}}$

no mass, no height: a property of the impact

NUMBERS

a check

drops 2 m, rebounds 1.28 m

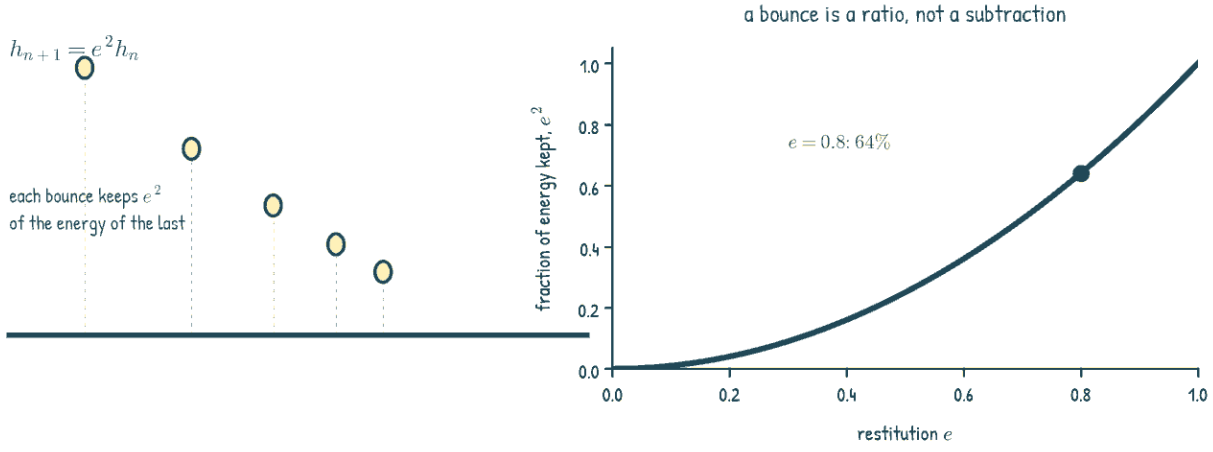
e = sqrt(1.28/2) = 0.8, energy kept 64 per cent

TRAP

e as an energy ratio

It is not. Energy scales as its square, so a rebound to half the height means e = 0.71, not 0.5.

restitution_energy



KEEP / e is the ratio of relative speeds, so energy is multiplied by e squared per impact

Who pays for the kinetic energy of a flywheel?

? A motor of constant power spins up a flywheel. The angular speed grows, but not linearly. Why, and who supplies the difference?

1 ## Power into rotation

The rotational kinetic energy is half I omega squared, so its derivative gives the power. That is the rotational twin of P = F v, with torque in place of force and omega in place of speed.

$$P = \frac{d}{dt}\left(\tfrac{1}{2} I\omega^2\right) = I\omega\dot{\omega} = \tau\omega$$

2 ## Integrate

With P constant the differential equation separates and gives omega proportional to the square root of time. The torque falls as one over omega, which is why a constant-power motor is slow at the start and cannot spin a wheel up evenly.

$$\omega = \sqrt{\frac{2Pt}{I}}, \qquad \tau = \frac{P}{\omega}$$

→ the same structure as v = sqrt(2Pt/m) for a car

3 ## Where the energy sits

All the work done by the motor ends up in the wheel: none is lost if the bearing is ideal. The question in the title is answered by the motor, and the only subtlety is the non-uniform rate at which it is delivered.

4 ## A worked flywheel

A flywheel of I = 8 kg m² is spun by a 300 W motor from rest. Find omega at 6 s and the time to 40 rad/s.

$$\omega = \sqrt{2(300)(6)/8} = 21.2 \text{ rad/s}$$

$$t = \frac{8(1600)}{2(300)} = 21.3 \text{ s}$$

→ the last 20 rad/s takes three times as long as the first 20

KEY RESULT

constant-power spin-up

$$\omega = \sqrt{2Pt/I}$$

and torque = P/omega

NUMBERS

a check

I = 12 kg m², P = 500 W

omega(10 s) = sqrt(833) = 28.9 rad/s

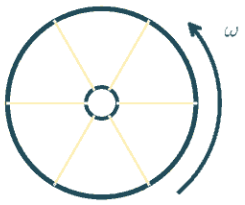
TRAP

using tau = I alpha with constant tau

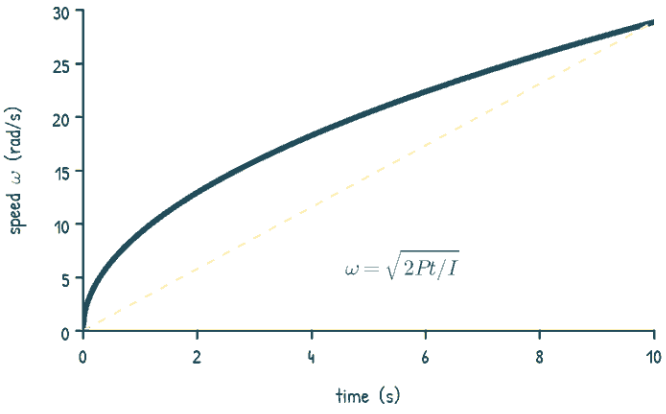
Constant power is not constant torque. Set tau = P/omega and solve the differential equation.

flywheel

a constant-power motor cannot spin up a wheel linearly



power divided by torque is angular speed



KEEP / at constant power the torque falls as one over omega, so the spin-up slows as root t

Can a force that is not there do work?

? A ball hangs in an accelerating cart. In the cart frame it swings backward and gains speed. Which force did the work?

1 ## Two frames, two accounts

In the ground frame the tension does the work that changes the ball speed; the pseudo force does not exist. In the cart frame the ball moves backward along the pseudo force, which does positive work equal to m a times that displacement.

$$W_{\text{pseudo}} = \int (-m\mathbf{a}_0) \cdot d\mathbf{r}$$

2 ## Why the frame version still balances

In a uniformly accelerating frame the pseudo force is constant and uniform, so it has a potential m a dot r and the energy book closes. In a rotating frame the pseudo forces include Coriolis, which is velocity-dependent and does no work, and centrifugal, which does and has a potential.

$$U_{\text{cf}} = -\tfrac{1}{2} m\omega^2 r_\perp^2$$

→ the bead on the hoop is the standard consumer of that potential

3 ## The exam rule

You may use energy in an accelerating frame only if you include the pseudo potential. Most students forget it and get a speed that is too large. The safe route is to compute energy in the inertial frame and forces in whichever frame is easier.

4 ## A worked frame

A 0.2 kg ball hangs in a cart accelerating at 4 m/s². It swings 0.25 m backward relative to the cart. Find the pseudo work and the ball speed in the cart frame.

$$W_{\text{pseudo}} = m a_0 d = 0.2(4)(0.25) = 0.20 \text{ J}$$

$$v = \sqrt{2(0.20)/0.2} = 1.41 \text{ m/s}$$

→ in the ground frame the tension did that work instead

KEY RESULT

pseudo potential

$$U_{\text{pseudo}} = -m\mathbf{a}_0 \cdot \mathbf{r}$$

centrifugal: -half m omega squared r squared

TRAP

Coriolis in an energy equation

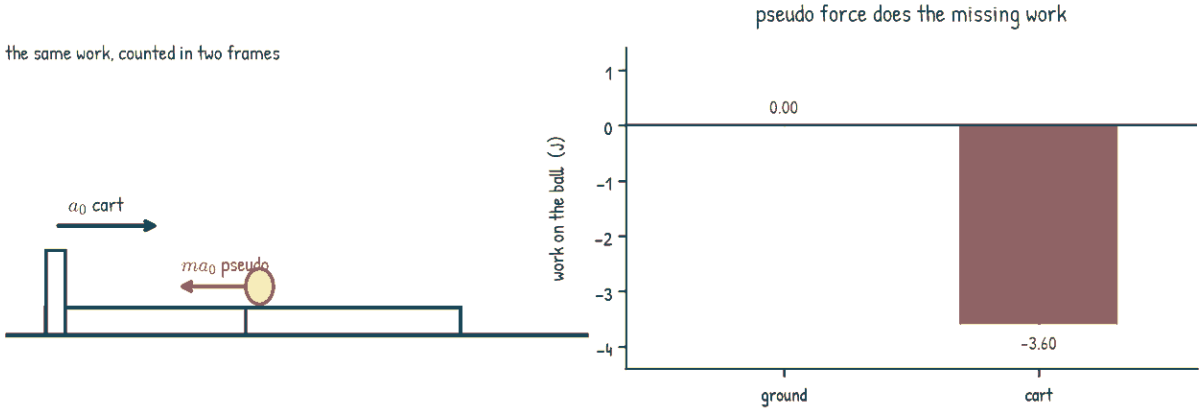
Coriolis is perpendicular to the velocity, so it does no work and has no potential. Only the uniform and centrifugal parts enter energy.

INSIGHT

the safe habit

Energy in the inertial frame, forces in the convenient frame. Mixing the two is where signs go wrong.

pseudo_work



KEEP / in an accelerating frame the pseudo force does work, and the frame energy book stays closed

Is one closed loop enough to prove a force is conservative?

? You are handed a force field and one closed loop. If the loop integral vanishes, may you conclude that a potential exists?

1 ## What the definition demands

Conservatism is a statement about all paths between any two points. A single loop that happens to close is a single data point; the field could still fail elsewhere.

$$\oint \mathbf{F} \cdot d\mathbf{r} = 0 \text{ for every loop}$$

2 ## The local test

For a force field in three dimensions the loop condition is equivalent to the curl vanishing everywhere in a simply connected region. In two dimensions that is the single condition that the cross derivative matches.

$$\nabla \times \mathbf{F} = 0 \quad \Longleftrightarrow \quad \partial_x F_y = \partial_y F_x$$

→ a vortex field passes no loop, and its curl is not zero

3 ## Why this matters in mechanics

Every potential energy you use in this chapter was licensed by that test. Gravity, the spring and the uniform field pass; friction and drag fail because they depend on the direction of travel or on speed, so no function of position can reproduce them.

4 ## A worked curl

Test F = (2xy, x²) N for conservatism, and find its potential.

$$\partial_x F_y = 2x = \partial_y F_x \text{ so the curl vanishes}$$

$$U = -x^2 y + \text{const}$$

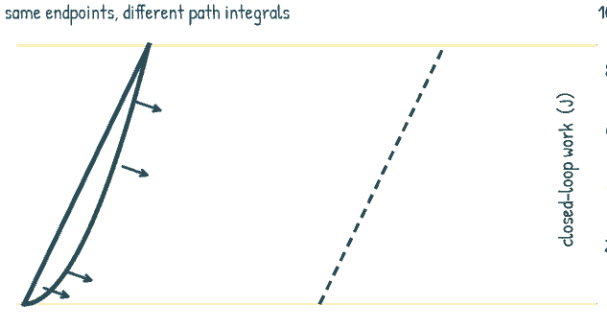
→ a vanishing curl plus a simply connected region is the licence for a potential

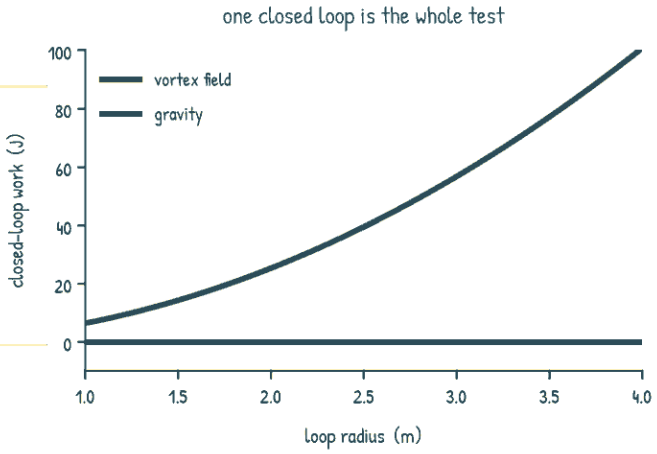
GATE
the two tests
global: all loop integrals zero. local: curl zero in a simply connected region.

NUMBERS
a check
F = (−y, x): curl = 2, loop work = 2 x area
nonzero for every loop: not conservative

TRAP
one loop as proof
A vanishing loop integral on a convenient circle proves nothing about the rest of the plane.

path_integral





KEEP / one loop proves nothing; every loop must vanish, which is what the curl tests

Why does doubling speed need eight times the power?

? A cyclist at 20 km/h is comfortable. At 40 km/h the same rider is exhausted. Why is the cost not twice?

1 ## The force law

At everyday speeds the drag force is proportional to the square of the speed. The power needed to overcome it is force times speed, so it goes as the cube.

$$f = bv^2, \qquad P = bv^3$$

2 ## The top speed

At constant speed the drive equals the drag, so a fixed power sets a maximum speed proportional to the cube root of the power. Four times the power buys only 1.6 times the speed, which is why the land speed record is so expensive.

$$v_{\max} = \left(\frac{P}{b}\right)^{1/3}$$

→ the cube root is the sting: 100 kW for a car means about 55 m/s with b = 0.6

3 ## The approach to the top

With a constant driving force the speed approaches terminal speed exponentially, and with constant power it approaches the ceiling from below without ever reaching it. Either way the last few metres per second cost most of the time.

$$v(t) = v_\infty(1 - e^{-t/\tau}) \quad \text{(constant drive)}$$

4 ## A worked drag

A cyclist and bike of 85 kg push 250 W. With b = 0.32, find the top speed on the flat.

$$250 = 0.32v^3 \;\Rightarrow\; v = (781)^{1/3} = 9.2 \text{ m/s}$$

$$\text{at } 500 \text{ W}: \; v = 11.6 \text{ m/s}$$

→ twice the power buys 26 per cent more speed, which is the whole story of cycling

KEY RESULT

drag power

$$f = bv^2, \qquad P = bv^3$$

NUMBERS

a check

b = 0.6, P = 100 kW

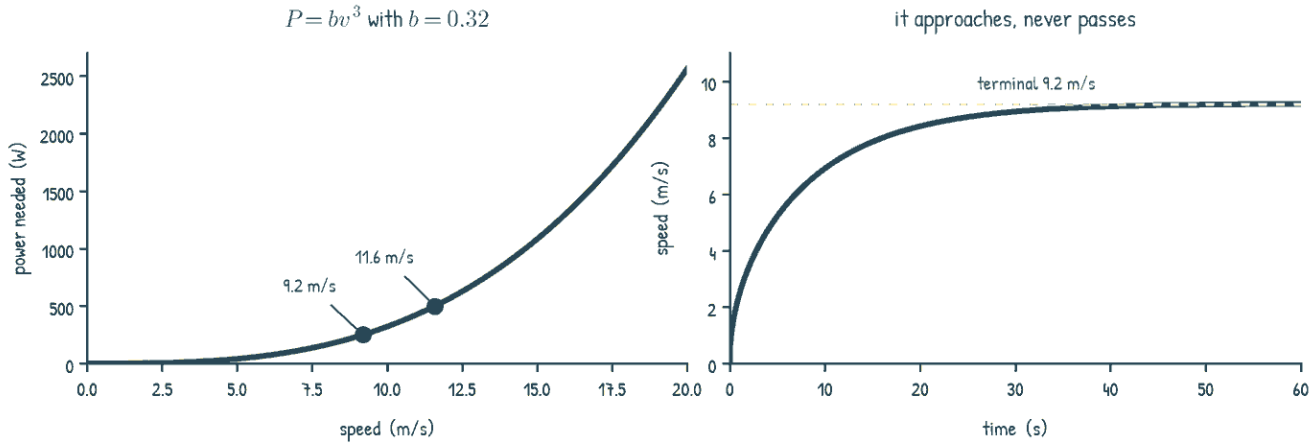
v_max = (100000/0.6)^(1/3) = 55 m/s

TRAP

linear scaling

Doubling speed quadruples the force and multiplies the power by eight. Linear intuition is wrong here by a factor of eight.

drag_power



KEEP / air resistance grows as the square of speed, and power as its cube

Where does the energy bookkeeping stop working?

? The chapter treats mechanical energy as exact. At what speed, and at what scale, does the ledger need new terms?

1 **Two fences**

Special relativity replaces half m v squared by (gamma minus one) m c squared once v is a noticeable fraction of c. Quantum mechanics makes the energy spectrum discrete once the action is comparable to h bar. Inside both fences the classical ledger is exact.

$$K = (\gamma - 1)mc^2 \approx \tfrac{1}{2}mv^2 \text{ for } v \ll c$$

2 **What survives**

The work-energy theorem survives both revolutions: work still equals the change in kinetic energy, with the relativistic kinetic energy. What does not survive is the formula half m v squared, and with it every classical estimate above about a tenth of the speed of light.

$$W = \Delta K \text{ always;} \quad K = \tfrac{1}{2}mv^2 \text{ only at low speed}$$

→ the theorem is more general than the formula it was derived from

3 **The practical boundary**

For a car, a satellite, a bullet, even a spacecraft at 10 km/s, gamma differs from one by less than one part in ten million. The relativistic correction is worth computing for electrons, for particle accelerators, and for nothing in this chapter.

4 **A worked ratio**

An electron at 2.0e8 m/s: what fraction of its kinetic energy does the classical formula miss?

$$\gamma = 1.34, \qquad K = (\gamma - 1)mc^2 = 0.342(511) = 175 \text{ keV}$$

$$\tfrac{1}{2}mv^2 = 114 \text{ keV} \;:\; \text{classical is } 35\ \%\ \text{low}$$

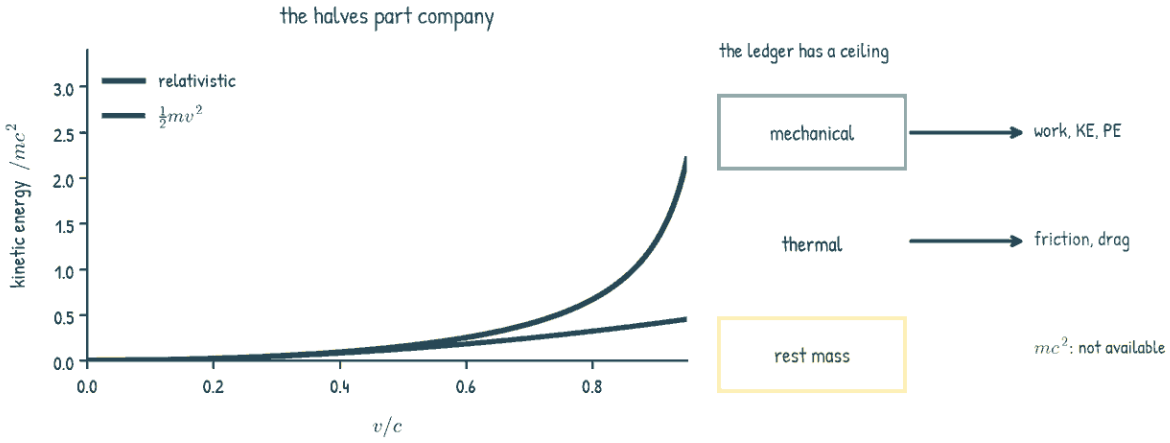
→ at v/c = 0.67 the classical formula is no longer an approximation, it is wrong

GATE
the ratio to check
Compute v/c. Below 0.1, classical is safe to a fraction of a per cent.

TRAP
quoting five figures
With g given to two figures and friction coefficients guessed, four-figure energy answers are a claim the data cannot support.

INSIGHT
what the chapter really teaches
Not the formulas, but the habit: name the store, name the loss, and check the ledger balances.

energy_limits



KEEP / the classical ledger is a low-speed approximation, and the rest mass joins it at high speed

What if the speed changes as well as the direction?

? A car on a curve brakes while turning. What is the total acceleration, and at what angle does it point?

1 ## Split the acceleration

Resolve along the tangent and the normal. The tangential part is the rate of change of speed; the normal part is v squared over R. They are perpendicular, so the magnitudes add in quadrature.

$$a_t = \frac{dv}{dt}, \qquad a_n = \frac{v^2}{R}, \qquad a = \sqrt{a_t^2 + a_n^2}$$

2 ## The direction is not radial

When a tangential component exists the total acceleration is tilted away from the centre, by an angle whose tangent is a_t over a_n. Only at constant speed does it point exactly at the centre.

$$\tan\phi = \frac{a_t}{a_n}$$

→ braking on a curve means the acceleration points backward and inward

3 ## The reverse question

Given the total acceleration and one component, recover the other and then the radius. This is the standard numerical form: a car has 13 m/s squared total and 5 m/s squared tangential at 10 m/s, so the normal part is 12 and the radius is 8.3 m.

$$a_n = \sqrt{a^2 - a_t^2}, \qquad R = \frac{v^2}{a_n}$$

4 ## A worked curve

A car at 18 m/s brakes at 3 m/s² while on a 40 m radius curve. Find the total acceleration and its angle from the radius.

$$a_n = \frac{324}{40} = 8.1\ \text{m/s}^2, \qquad a = \sqrt{9 + 65.6} = 8.64\ \text{m/s}^2$$

$$\phi = \tan^{-1}\frac{3}{8.1} = 20.3^\circ$$

→ the car is slowing and turning at once, and the vector says which dominates

KEY RESULT

the two components

$$a_t = \dot{v}, \qquad a_n = v^2/R$$

NUMBERS

a check

v = 10, a_t = 5, a = 13

a_n = 12, R = 8.3 m

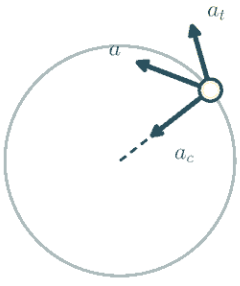
TRAP

using R from the total acceleration

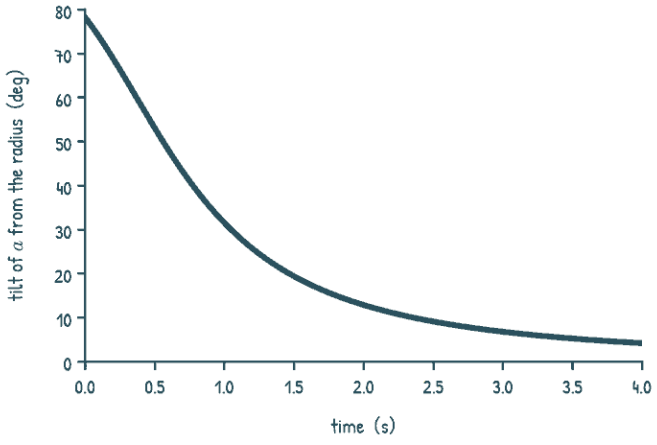
Only the normal component sets the radius. The tangential part does not bend the path at all.

at_an

$$a = \sqrt{a_t^2 + a_c^2}$$



as it speeds up, a turns radial



KEEP / two perpendicular components: one changes the speed, one changes the direction

What if the angular acceleration is not constant?

? A wheel slows under a torque that falls with time. The three rotation equations fail. What replaces them?

1 Back to the definitions

When alpha varies, go back to the derivatives. Integrating alpha over time gives the change in omega, and integrating omega gives the change in angle. Each integral is an area under a graph.

$$\omega = \omega_0 + \int_0^t \alpha\, dt, \qquad \theta = \theta_0 + \int_0^t \omega\, dt$$

2 The other chain

If alpha is given as a function of the angle instead of the time, use the rotational twin of v dv/dx. It removes the clock and often turns a hard integral into an easy one.

$$\omega\, d\omega = \alpha\, d\theta \quad \Rightarrow \quad \tfrac{1}{2}\omega^2 = \int \alpha\, d\theta$$

→ the same trick as a = v dv/dx, in rotation

3 A worked shape

For a wheel slowing under friction with alpha proportional to minus omega, the solution is exponential decay of the angular speed: each second removes the same fraction of what is left. That is the rotational analogue of linear drag.

$$\alpha = -k\omega \;\Rightarrow\; \omega = \omega_0 e^{-kt}$$

4 A worked decay

A wheel slows with alpha = −0.8 omega rad/s². From 25 rad/s, find omega at 2 s and the total angle before stopping.

$$\omega = 25e^{-1.6} = 5.05 \text{ rad/s}$$

$$\theta = \frac{\omega_0}{k} = \frac{25}{0.8} = 31.3 \text{ rad} = 5.0 \text{ turns}$$

→ an exponential spin-down never quite stops, but the angle converges

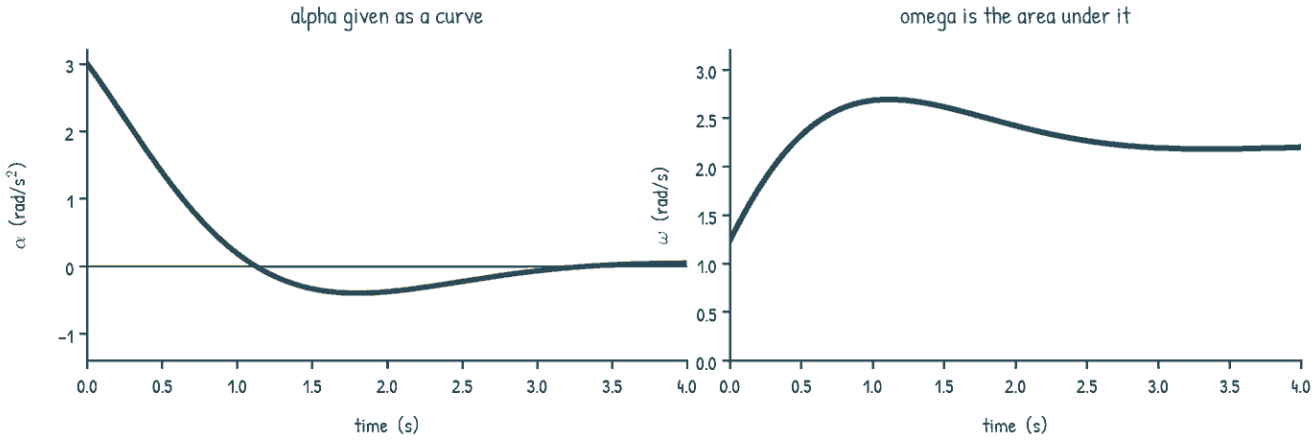
KEY RESULT
the integrals

$$\omega = \int \alpha\, dt, \qquad \theta = \int \omega\, dt$$

GATE
choose the variable
alpha of time: integrate over t. alpha of angle: use omega d omega = alpha d theta.

TRAP
reusing omega_0 + alpha t
That equation assumes constant alpha. Check before writing it, and if alpha is a function, integrate.

alpha_integral



KEEP / integrate: omega is the area under alpha, theta the area under omega

How fast do two points on a wheel separate?

? Two particles move on the same circle at different angular rates. How often do they meet, and what is their relative speed?

1 ## Subtract the rates

Both particles share the same circle, so their separation angle changes at the difference of the angular velocities. Meetings happen once per full relative revolution, which gives a period of 2 pi over the difference.

$$\omega_{\text{rel}} = \omega_1 - \omega_2, \qquad T_{\text{meet}} = \frac{2\pi}{|\omega_1 - \omega_2|}$$

2 ## Relative speed along the circle

The separation grows along the arc, so the arc grows at the relative angular rate times the radius. For two independent particles on one circle this is the closing speed along the track; for two points of one rigid wheel the rates are equal and the gap never changes.

$$v_{\text{rel}} = R\,|\omega_1 - \omega_2|$$

→ a rigid wheel spins its points together: a shared rate means a fixed gap

3 ## The same idea across frames

If the whole circle is also rotating, add or subtract the frame rate first. This is the rotational version of relative velocity, and it is what makes gear trains and pulleys tractable: the contact points share a velocity, not a rate.

4 ## A worked meeting

Two runners on a 200 m track go at 6 and 4.5 m/s the same way. The track length, not the radius, is given. When do they next meet?

$$R = \frac{200}{2\pi} = 31.8\ \text{m}, \quad \omega_1 = 0.188,\ \omega_2 = 0.141\ \text{rad/s}$$

$$T_{\text{meet}} = \frac{2\pi}{0.0471} = 133\ \text{s} = 2.2\ \text{min}$$

→ opposite ways the rates add: they then meet every 19 s

KEY RESULT

relative rotation

$$T_{\text{meet}} = 2\pi/|\omega_1 - \omega_2|$$

NUMBERS

a check

omega = 1.5 and 0.6 rad/s

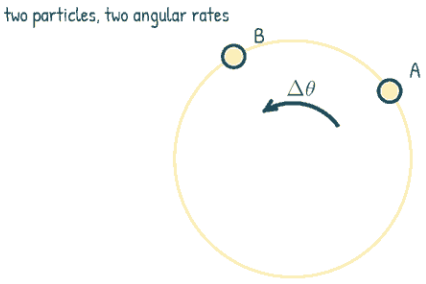
T_meet = 2 pi / 0.9 = 7.0 s

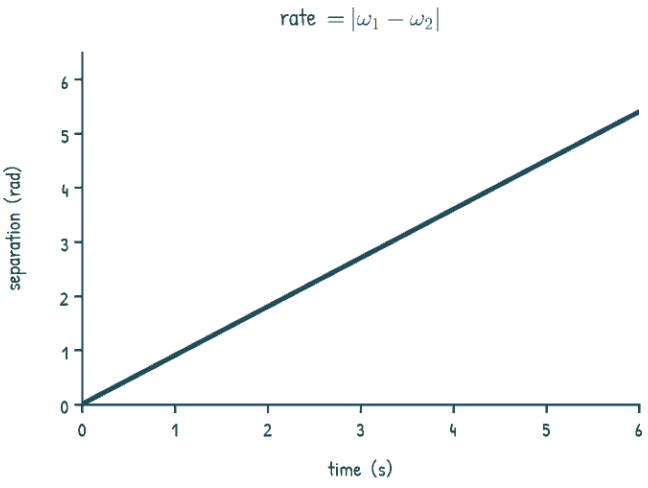
TRAP

adding the rates

Same-direction rotation subtracts. Adding gives the answer for opposite directions.

relative_angular





KEEP / relative angular rate is the difference, and relative speed is that times the radius

If the centripetal force is always inward, how does anything speed up?

? On a vertical circle gravity has a tangential component; the tension never does. Which force changes the speed, and which only bends the path?

1 ## Perpendicular means zero work

The centripetal force is along the radius and the displacement is along the tangent, so the dot product vanishes at every instant. It therefore cannot change the kinetic energy, which is why a car on a flat curve keeps a constant speed while its velocity turns.

$$W_c = \int \mathbf{F}_c \cdot d\mathbf{r} = 0$$

2 ## Who speeds the body up

Only the tangential component does work. On a vertical circle that is the tangential part of gravity, m g sin of the angle from the lowest point, so the work done by gravity is exactly minus the change in m g h. On a banked curve it is friction along the path.

$$m\tfrac{dv}{dt} = F_{\text{tangential}} = mg\sin\phi$$

3 ## Why the two equations are separate

The radial equation fixes the force needed for the instantaneous geometry; the tangential equation fixes how the speed evolves. Solving them in that order is the standard approach, and it is the reason energy can replace the tangential equation entirely.

→ energy handles the tangential direction; the radial equation handles the normal one

4 ## A worked split

A 0.5 kg bead on a 0.6 m string is at 4 m/s when the string makes 60° with the vertical, moving up. Find the tangential acceleration and the tension.

$$a_t = g\sin 60^\circ = 8.49\ \text{m/s}^2$$

$$T = \tfrac{0.5(16)}{0.6} + 0.5(9.8)\cos 60^\circ = 13.3 + 2.45 = 15.8\ \text{N}$$

→ gravity does the speeding up or slowing down; the string only bends the path

KEY RESULT

the split

radial: sets the path, no work. tangential: sets the speed, does the work.

TRAP

tension doing work

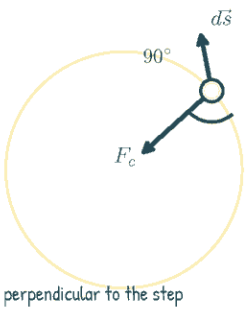
For a string of fixed length the tension is perpendicular to the motion and does no work. A string that changes length, or a peg that moves, can.

INSIGHT

the energy shortcut

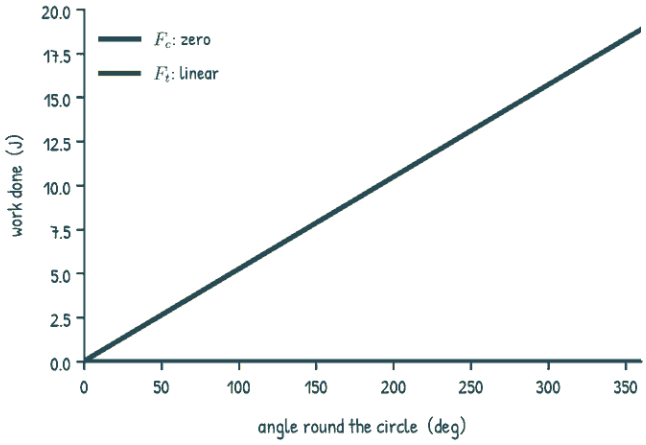
Because the radial force is powerless, the whole tangential problem collapses to conservation of mechanical energy.

work_radial



the radial force is always perpendicular to the step

only the tangential part earns work



KEEP / a radial force does no work, so only the tangential component can change the speed

What speed does a bead have at the top of a loop?

? A bead is released from rest at height h above the bottom of a smooth vertical loop. How fast is it at the top, and when does it leave the track?

1 Energy from the release point

With no friction the speed at any angle follows from the height dropped. Measuring the angle from the bottom, the height gained is R times one minus cosine, and the speed follows directly.

$$v^2 = v_0^2 - 2gR(1-\cos\phi)$$

2 The track condition

For a bead on the inside of a track the normal must stay positive. Setting it to zero gives the minimum speed at each angle, and the most demanding point is the top, where gravity already supplies the whole centripetal requirement.

$$N = \frac{mv^2}{R} + mg\cos\phi \;\geq 0$$

→ at the top cos phi = −1, so N = m v squared / R − m g

3 The two thresholds

For a full loop without leaving the track the top speed must satisfy v squared at least g R, which needs a release height of two and a half radii above the bottom. For a bead on a rod, or a bead threaded on a wire, there is no such condition and the minimum is only the height of the top.

$$h_{\min} = 2.5R \text{ (track)}, \qquad h_{\min} = 2R \text{ (rod)}$$

4 A worked loop

A 0.2 kg bead is released from rest 1.5 m above the bottom of a 0.5 m loop. Find the speed at the top and say whether it stays on.

$$v^2 = 2(9.8)(1.5-1.0) = 9.8 \;\Rightarrow\; v = 3.13 \text{ m/s}$$

$$v_{\text{top, min}}^2 = gR = 4.9 \;:\; 9.8 > 4.9, \text{ it stays on}$$

→ the margin is exactly a factor of two in v squared

KEY RESULT

the pair

$$v^2 = v_0^2 - 2gR(1-\cos\phi)$$

$$N = \frac{mv^2}{R} + mg\cos\phi$$

NUMBERS

a check

R = 0.8 m, released from rest at 2.5R

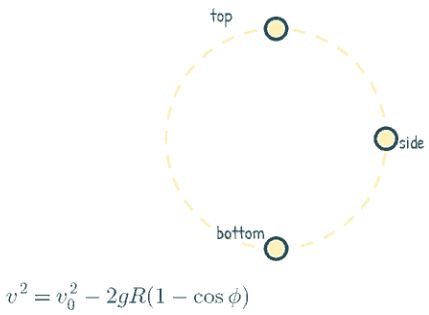
v_top = sqrt(gR) = 2.8 m/s, N = 0 there

TRAP

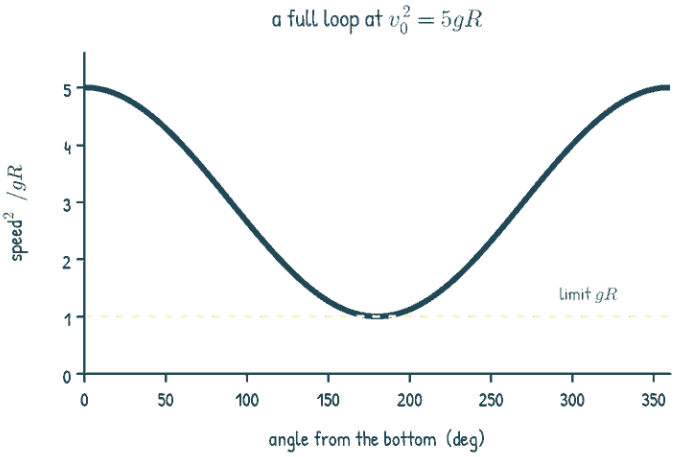
using v = 0 at the top

A bead on the inside of a track needs sqrt(gR) at the top, not zero. Zero is the rod answer.

vertical_energy



$v^2 = v_0^2 - 2gR(1-\cos\phi)$



KEEP / energy gives the speed at every angle; the track condition decides whether it stays on

Why must a string be launched faster than a rod?

? The same mass, radius and release height. On a string it must be launched at sqrt(5gR); on a rod, sqrt(4gR) is enough. Where does the difference come from?

1

The constraint is one-sided

A string can pull but not push, so the tension must remain non-negative everywhere. A rod can supply either sign, so the only requirement is that the bead reaches the top with enough speed to keep going.

$$T \geq 0 \text{ (string)}, \qquad N \text{ any sign (rod)}$$

2

Both boundary cases

At the top of a string loop the tension just vanishes at the limiting speed, giving v squared equal to g R there and 5 g R at the bottom by energy. For a rod the limiting speed at the top is zero, which gives 4 g R at the bottom.

$$v_{\text{bot}} = \sqrt{5gR} \text{ (string)}, \qquad \sqrt{4gR} \text{ (rod)}$$

→ the factor sqrt(5/4) = 1.12 in speed is a factor 1.25 in energy

3

What the tension does along the way

For the string at the limiting launch speed the tension is zero at the top and 6 m g at the bottom. For the rod the normal force changes sign somewhere above the horizontal, and the bead is being pushed inward there rather than pulled.

$$T_{\text{bot}} = 6mg \text{ (string, limiting)}, \qquad N_{\text{top}} = -mg \text{ (rod, limiting)}$$

4

A worked comparison

A 0.2 kg ball on a 0.9 m string is whirled in a vertical circle. Find the minimum bottom speed and the tension there.

$$v_{\text{bot}} = \sqrt{5(9.8)(0.9)} = 6.64 \text{ m/s}$$

$$T_{\text{bot}} = 6mg = 11.8 \text{ N}$$

→ a rod would need only 5.94 m/s, and a bottom force of 5mg

KEY RESULT

the two minima

$$v_{\text{bot}} \geq \sqrt{5gR} \text{ (string)}, \qquad \geq \sqrt{4gR} \text{ (rod)}$$

NUMBERS

a check

R = 0.8 m: string 6.26 m/s, rod 4.85 m/s

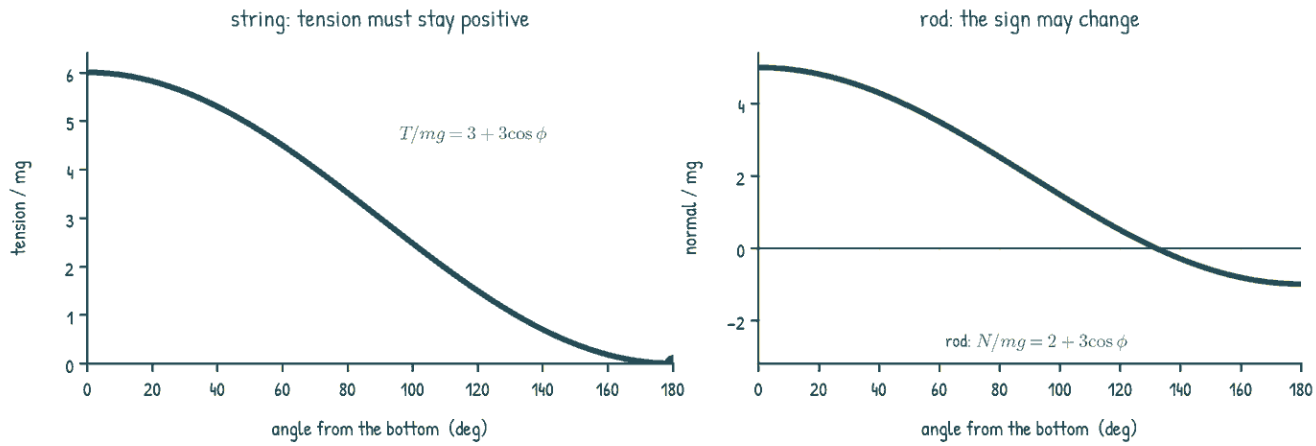
tensions at the bottom: 6mg and 5mg for the two limits

TRAP

the source 3mg

The bottom tension at the limiting speed is 6mg, not 3mg. See the source audit for the corrected figure.

string_vs_rod



KEEP / a string can only pull, so its tension must stay positive; a rod can push

Why does a conical pendulum ignore the mass?

? A bob whirls in a horizontal circle on a string of length L at angle theta. The tension depends on the mass, yet the period does not. Why?

1 ## Two equations at once

Vertically the vertical component of the tension balances the weight. Horizontally the horizontal component supplies the centripetal force. Divide one by the other and the mass and the tension both disappear.

$$T\cos\theta = mg, \qquad T\sin\theta = \frac{mv^2}{L\sin\theta}$$

2 ## What survives

The division leaves the angle fixed by the speed, and the period depends only on the vertical length of the string. Every bob on the same string length and the same angle has the same period, whatever its mass.

$$v^2 = gL\sin\theta\tan\theta, \qquad T = 2\pi\sqrt{\frac{L\cos\theta}{g}}$$

→ the vertical projection L cos theta is the length that matters

3 ## The limits

As the angle goes to zero the motion becomes a small planar pendulum with period 2 pi root L over g. As the angle approaches 90 degrees the required speed diverges and the period falls to zero: the string becomes horizontal only at infinite speed.

$$\theta \to 0:\ T \to 2\pi\sqrt{L/g}, \qquad \theta \to 90^\circ:\ v \to \infty$$

4 ## A worked whirl

A 1.2 m string carries a 0.3 kg bob at 30° to the vertical. Find the tension, speed and period.

$$T = \frac{0.3(9.8)}{\cos 30^\circ} = 3.40\ \text{N}$$

$$v = \sqrt{9.8(1.2)\sin 30^\circ \tan 30^\circ} = 1.84\ \text{m/s}$$

$$T_{\text{period}} = 2\pi\sqrt{1.2(0.866)/9.8} = 2.05\ \text{s}$$

→ the same string at 60° would need 2.4 times the speed and swing faster

KEY RESULT

the conical pendulum

$$T = 2\pi\sqrt{L\cos\theta/g}, \qquad v = \sqrt{gL\sin\theta\tan\theta}$$

NUMBERS

a check

L = 0.5 m, theta = 40°, m = 0.2 kg

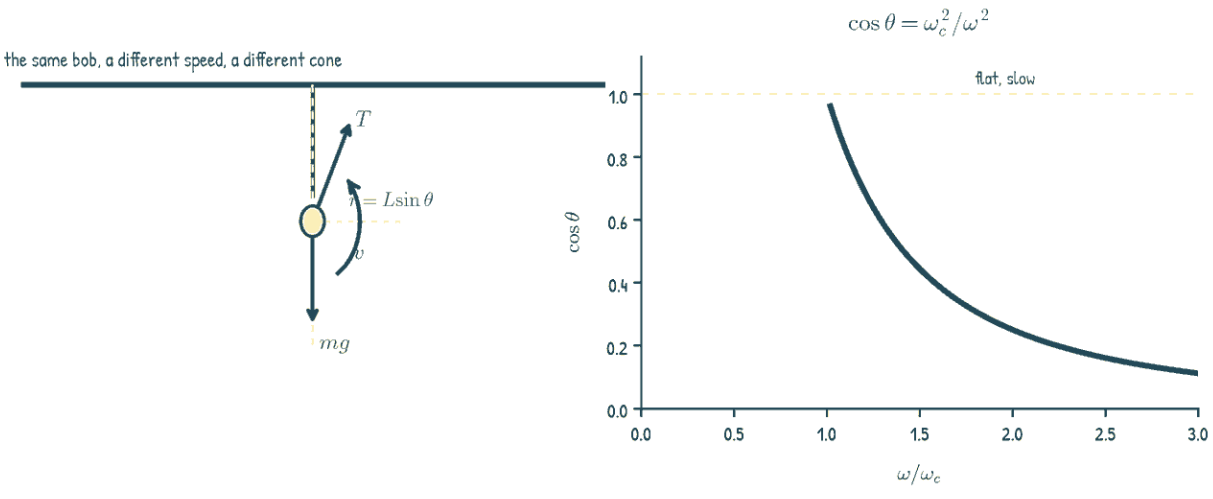
T = 1.24 s, v = 1.63 m/s, tension 2.56 N

TRAP

using L in the period

The period uses L cos theta, the vertical drop, not the string length.

conical_energy



KEEP / the two force equations both carry m, so it cancels out of the geometry and the period

What is the slowest safe speed on a banked corner?

? Banking is designed for one speed. Too fast and the car slides up; too slow and it slides down. What sets both limits?

1 ## Design speed

With no friction the normal force alone must supply the centripetal force, and resolving along and perpendicular to the bank gives the design condition.

$$\tan\theta = \frac{v_0^2}{Rg}$$

2 ## The two extremes

At the maximum speed friction points down the slope, helping the bank turn the car. At the minimum it points up the slope, stopping the car sliding down. Both come from the same resolution with the sign of friction flipped.

$$v_{\max}^2 = Rg\frac{\sin\theta + \mu\cos\theta}{\cos\theta - \mu\sin\theta}$$

$$v_{\min}^2 = Rg\frac{\sin\theta - \mu\cos\theta}{\cos\theta + \mu\sin\theta}$$

3 ## Read the structure

Both reduce to tan theta when mu vanishes, the design speed. The maximum grows without bound as mu approaches cot theta, while the minimum falls to zero once mu exceeds tan theta.

$$\mu > \tan\theta \;\Rightarrow\; v_{\min} = 0$$

→ the band is asymmetric: friction helps much more at the fast end

4 ## A worked band

A 120 m radius corner is banked at 15° with mu_s = 0.45. Find both speed limits.

$$v_{\max}^2 = 1176\frac{0.259 + 0.435}{0.966 - 0.117} = 963 \;\Rightarrow\; 31.0\ \text{m/s}$$

$$v_{\min}^2 = 1176\frac{0.259 - 0.435}{0.966 + 0.117} < 0 \;\Rightarrow\; v_{\min} = 0$$

→ mu exceeds tan 15° = 0.268, so the car can stand still on the bank

KEY RESULT

the band

$$v_{\max}^2 = Rg\frac{\sin\theta + \mu\cos\theta}{\cos\theta - \mu\sin\theta}$$

and the min with the signs flipped

NUMBERS

a check

R = 80, theta = 27°, mu = 0.35

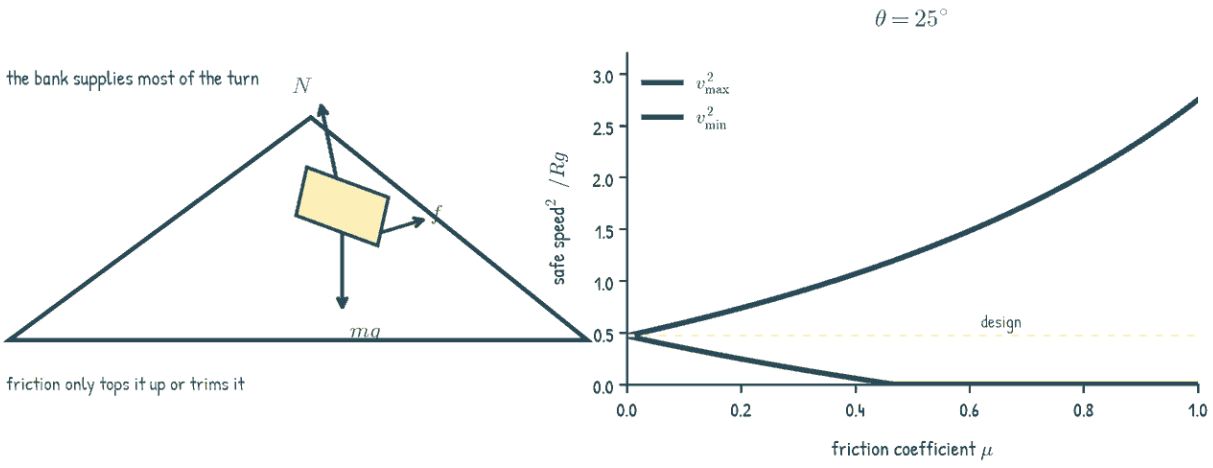
v: 10.3 to 28.6 m/s, design 20 m/s

TRAP

one formula for both

The friction direction flips at the design speed. Using one form for both limits gives a wrong minimum.

bank_limits



KEEP / friction can only widen the band around the design speed; it cannot create one

How does a wall of death hold a rider up?

? A rider circles the inside of a vertical cylinder. The wall is vertical and there is no floor under the tyres. What holds the weight?

1

Two perpendicular jobs

The normal force is horizontal, points at the axis, and provides the centripetal force. Friction is vertical and holds the weight. Neither can do the other job, so the two equations are independent.

$$N = \frac{mv^2}{R}, \qquad f = mg$$

2

The minimum speed

Static friction has a ceiling proportional to the normal. Substituting the two equations gives the minimum speed at which the wall can still hold the rider, and it falls as the friction coefficient grows.

$$\mu_s \frac{mv^2}{R} \geq mg \quad \Rightarrow \quad v_{\min} = \sqrt{\frac{gR}{\mu_s}}$$

→ a smaller cylinder, or a stickier wall, lowers the required speed

3

Why the rider feels heavy

At the minimum speed the friction is exactly at its limit, so any bump or lean that reduces the normal drops the rider. That is why the manoeuvre is done at well above the minimum, and why the normal force on the rider can be several times the weight.

4

A worked rotor

A 70 kg rider circles a 5 m radius wall of death at mu_s = 0.6. Find the minimum speed and the normal force then.

$$v_{\min} = \sqrt{9.8(5)/0.6} = 9.04 \text{ m/s}$$

$$N = \frac{70(81.7)}{5} = 1.14 \text{ kN}, \qquad f = 686 \text{ N}$$

→ at the minimum speed the friction is exactly at its ceiling

KEY RESULT

the rotor

$$v_{\min} = \sqrt{gR/\mu_s}$$

normal horizontal, friction vertical

NUMBERS

a check

R = 4 m, mu_s = 0.4

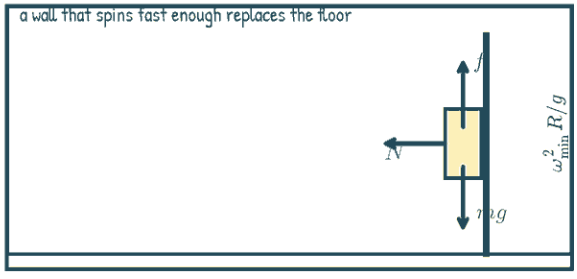
v_min = sqrt(98) = 9.9 m/s, about 36 km/h

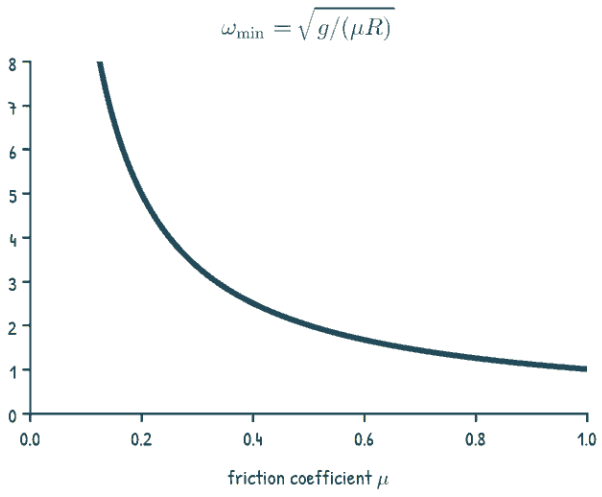
TRAP

friction supplying the centre

On a vertical wall friction cannot point at the axis. The normal does that; friction is vertical.

rotor_wall





KEEP / the normal provides the centripetal force, and static friction on the vertical wall carries the weight

How tight is a curve that is not a circle?

? A projectile path, a roller-coaster dip, a road bend: none is a circle. What is the radius at a point?

1 Define it by the normal acceleration

At any point of any smooth path there is one circle that matches it to second order. Its radius is the radius of curvature, and the normal component of acceleration is v squared over that radius.

$$\rho = \frac{v^2}{a_\perp}$$

2 Projectile values worth knowing

At the apex the velocity is horizontal and the whole of g is normal, so the radius is v squared over g with the horizontal speed. At launch the normal component is g cos theta, which gives a larger radius.

$$\rho_{\text{apex}} = \frac{v_0^2 \cos^2\theta}{g}, \qquad \rho_{\text{launch}} = \frac{v_0^2}{g\cos\theta}$$

→ for a vertical launch the path is straight and the radius is infinite

3 The general formula

For a path given as y of x, the radius follows from the first and second derivatives. A straight line has zero second derivative and infinite radius; a sharp bend has a large second derivative and a small radius.

$$\rho = \frac{(1 + y_x^2)^{3/2}}{|y_{xx}|}$$

→ at the apex of a parabola y double dash is g over v_x squared, which reproduces the result above

4 A worked curvature

A projectile is launched at 25 m/s at 40°. Find the radius of curvature at the apex and at launch.

$$\rho_{\text{apex}} = \frac{625\cos^2 40^\circ}{9.8} = 37.3 \text{ m}$$

$$\rho_{\text{launch}} = \frac{625}{9.8\cos 40^\circ} = 83.2 \text{ m}$$

→ the path bends hardest where it is least steep and slowest in the horizontal

KEY RESULT

curvature

$$\rho = v^2 / a_\perp$$

and rho = (1 + y_x^2)^(3/2) / |y_xx| for a graph

NUMBERS

a check

20 m/s at 45°: apex rho = 200/9.8 = 20.4 m

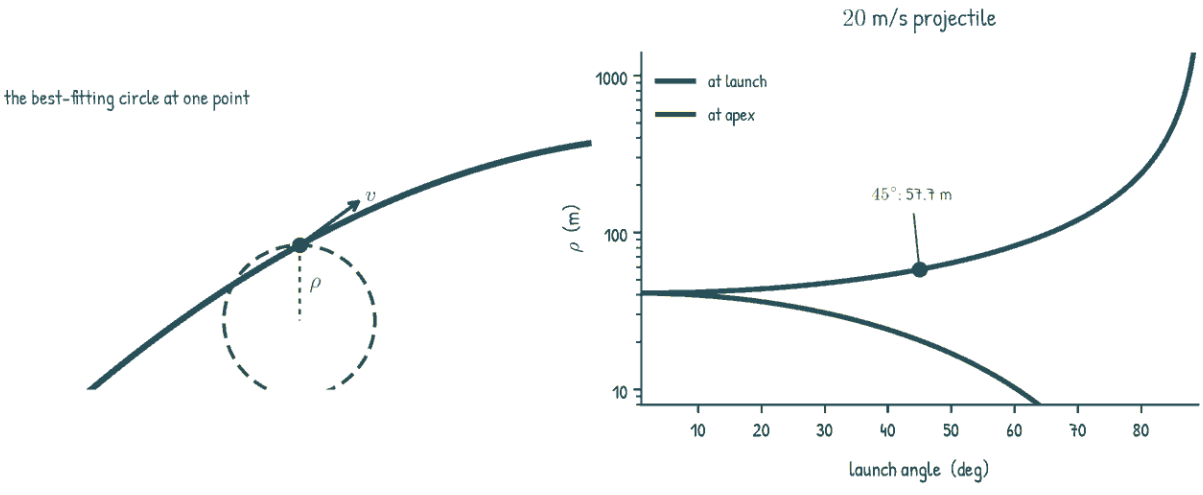
launch rho = 400/(9.8 x 0.707) = 57.7 m

TRAP

using the full speed at the apex

The apex speed is v0 cos theta, not v0. Using v0 overstates the radius by 1/cos² theta.

curvature_fit



KEEP / the radius of curvature is the circle that matches the path locally, and v squared over it is the normal acceleration